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Image Forgery Detection using Machine Learning with Fusion of Global and Local Thepade's SBTC Features

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***Abstract*—Image forgery is manipulating digital images to hide or change some useful information contained in the images. Images are considered the most effective way to convey information, and manipulating this information sometimes creates havoc. The action of tampering with images that are done either for fun or to give false evidence has resulted in a disaster in some cases. It is done in such a way that it cannot be determined by the naked human eye, so many people have implemented various types of machine learning algorithms, which they have implemented with handcrafted features to determine different types of forgery and whether an image is forged or not. These algorithms are used to extract the digital signature differentiating whether an image has been tampered with or not. Various techniques have been implemented for either fine or coarse image splicing, whereas a technique dealing with both needs to be devised. For this, our proposed work focuses on different types of machine learning classifiers and 10-fold classification. The attempted values of n for the machine learning classifier include 2,3,4. The different types of classifiers include Random Forest, Random tree, support vector machine, Logistic Regression, Naive Bayes. These classifier models are trained on comofod, casia v2.0 datasets. Accuracy increase is observed when a fusion of Thepade's Sorted Block Truncation Coding (i.e., Thepade's SBTC) local and Thepade's SBTC global feature tables.**

Keywords—image classification, image splicing, forgery detection, machine learning, classification, fake images, Thepade's SBTC feature extraction.

I. INTRODUCTION

With the cutting-edge computerized world and advanced innovation approach, billions of pictures are being produced, shared, or transferred day by day. With many photo editing software like Adobe, GIMP, and its great ease of use, it becomes very easy for anyone to tamper with an image. Millions of forged images are being uploaded daily, used for false evidence, hiding information. In courtrooms, images are used as evidence in visual form, and hence tampered images meaning tampered evidence can result in punishing an innocent person. Like this, forged images that float and spread on social media platforms spread fake news and fake information. Hence can harm an individual, can even result in rivalry and conflict among societies with diverse cultures. Recently various false beliefs about the covid19 pandemic were spread in the form of fake images, which created misbeliefs amongst citizens. [6] Here are examples of famous image forgery done to date. Thus, various tools need to be developed to detect the genuineness of the image. Many attempts have been made in this direction where people have used handcrafted features and implemented CNN, deep learning. The number of experiments is still going on.

Splicing could be fine and coarse. Although many attempts have been made in the direction of splicing detection, a technique dealing with fine and coarse equally versed efficiently needs to be devised, which is the proposed method in this paper.

Key Contribution of the work presented-

- Used global Thepade's SBTC 2,3,4 ary.
- Calculated the local Thepade's SBTC for 2,3,4 ary.
- Implemented fusion of local and global Thepade's SBTC as a digital signature.
- Calculated accuracy for various machine learning classifiers using ten-fold cross-validation.

Paper partitioning is done as follows: Section II comprises the literature survey, which contains the current philosophies. Section III gives the Proposed Method outline. Section IV explains the environment for experimentation. The outcomes, tables, and conversations are exhibited in Section V. Last comments are closed in Section VI.

II. LITERATURE SURVEY

In image forgery detection [11][12], two approaches followed are active and passive approaches. In the active approach, the focus has mainly been on pre-embedded

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information and pre-registration, which involves digital watermarking and digital signatures, while in the passive approach, there is no pre-information. Hence, passive image forgery detection is harder to detect and has become a hot topic for research. Copy-move forgery and image splicing detection are the types of passive approaches. In image splicing, duplication, and glueing are done from the same or various sources. This path consists of at least two pictures consolidated to make a phoney picture, while in copy-move forgery, a piece of one picture is reordered onto a similar picture.

The paper [13] presented work on content-based image retrieval (CBIR) strategies dependent on staggered Block truncation coding with the utilization of multiple threshold values. Block Truncation Coding-based components, one of the CBIR strategies proposed utilizing shading provisions of image. The methodology essentially considered red, green, and blue planes of an image together to include vectors. The averaging colour techniques utilized here were BTC Level-1, BTC Level-2, BTC Level-3. Here the component vector size per image has incredibly decreased by utilizing the mean of each plane and finding out the threshold value, then dividing each plane using a threshold value. Here, the proposer of the method applied colour averaging to calculate precision and recall to calculate the algorithm's performance.

In paper [7], image splicing recognition in chroma spaces has presented dependent on the RLRN characteristics, in this utilized SVM as a classifier. The analysts proposed that assuming discovery of grafting was troublesome in one shading space, it very well might be simpler with another shading space and consequently utilized a chroma space to extricate four run-length run numbers as feature vectors with various headings. The perception was that presentation of RLRN in the chroma channel (Cr or Cb) outperformed luma and RGB channels. The test was performed on two datasets: casia v1.0 and Columbia, where precision estimates were 82.1%, 85%, and 94.3%, 94.7% on Cr and Cb individually.

In paper [9] colour image splicing recognition technique has been proposed. The method utilized picture chroma, which reflects splicing introduced modification. This splicing was distinguished dependent on the grey level co-occurrence matrix that is GLCM characteristics of edge picture. The highest accuracy obtained following extraction of features resulted in 90.5%, which proved effective in detecting splicing based on image edge observation in the chroma channel.

Except for various machine learning techniques mentioned above for image forgery detection, existed numerous methods for detection of image forgery using a deep learning approach. With the advent of various pre-trained or deep learning architectures, many researchers used them for forgery detection.

In paper [8], the proposed approach detected forged images using a deep learning approach. The main objective of this approach overcame less availability of training data to fine-tune networks having complex architectures, for example, VGG-16. The classification strategy is based upon image fragments. Training and testing were done on a casia dataset, and classification precision of 97.8% was observed on a fine-tuned model and obtained 96.4% exactness for zero stage training.

The Thepade's Sorted Block Truncation Coding is widely being used for variety of application areas related to image and video classification [15, 16]

In paper [10], research work proposed a two-stage deep learning method to learn features to determine the tampered image over different image formats. They obtained an accuracy of 91.09% over both JPEG and TIFF for the CASIA dataset. For the JPEG tampered image, the accuracy was 87.51% which performs better from two state-of-the-art methods.

III. Methodology

A. Proposed Method

In the proposed strategy, a feature extraction technique for forgery detection, which is named 'Thepade's SBTC local feature extraction method' has been introduced. The results obtained with this are observed and noted in the following section. Then, Thepade's SBTC global n-ary method [1] for global feature extraction is combined with a proposed method, Thepade's SBTC local feature extraction, and results are obtained and noted in the following section.

Different alterations for global Thepade's SBTC are considered for finding the feature vectors. The colour planes of the images are extracted for R, G & B values. These are later converted into feature vectors used for classification using various machine learning algorithms to categorize the class of pictures as original or tampered.

The overall flow of experiment is demonstrated in figure 1. Feature extraction is done from Image dataset after which feature fusion is done. The classifier is then trained using fusion of all the features and final results are obtained.

The proposed method is demonstrated in figure 2. Extraction of TSBTC local and global features is done from Image dataset and feature fusion of these are performed. Fusion of these features are then fed to classifier. Here the feature vectors are extracted for various images available in the training dataset, including 2,3,4 ary of Thepade's SBTC. Six classifiers are used on which the feature vectors are validated to extract an optimal training model that can be used for image classification, namely Tampered and Original.

And finally, as shown in figure 3, query image, that is to be identified, is fed to trained machine learning classifier where it is classified as either original image or tampered image.

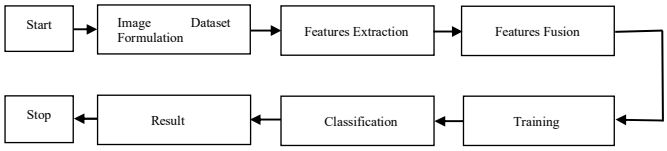

Fig. 1. The overall flow of the experiment.

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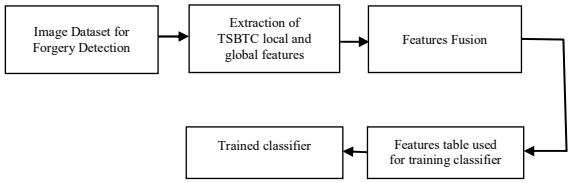

Fig. 2. Training of ML model.

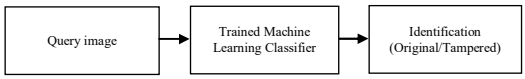

Fig. 3. Using ML model for image categorization.

a. Feature Extraction using global TSBTC n-ary

1. R, G, B colour planes are extracted from the picture.
2. These R, G, B pixel values are stored in three different vectors of each channel.
3. Sorting operations are performed on these vectors. Thus, three old unsorted vectors are replaced by three new sorted vectors.
4. For the ordered vectors Red, Green, Blue mean value of each part is evaluated by dividing it into n parts using the equation as below (a), (b), (c)

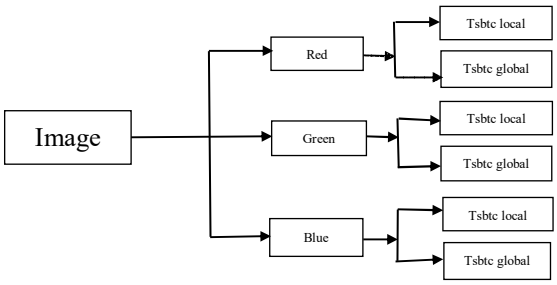

Fig. 4. Image colour planes extraction.

(a)

$$B1 = \frac{3}{k*p} * \sum_{c=1}^{\frac{k*p}{3}} \quad Sorted\ B$$

(b)

$$B2 = \frac{3}{k*p} * \sum_{c=\frac{k*p}{3}+1}^{\frac{2*k*p}{3}} \quad Sorted\ B$$

(c)

$$B3 = \frac{3}{k*p} * \sum_{c=\frac{2*k*p}{3}+1}^{k*p} \quad Sorted\ B$$

Similarly, Red and Green feature vectors are calculated as (R3, R1, R2) and (G3, G1, G2), respectively.

b. Feature Extraction for local TSBTC n-ary

1. Extracted the R, G & B colour planes from the given image.
2. Window size of 3*3 is considered where the pixel values for window size are extracted and stored in a vector for each colour plane.
3. These pixel values are sorted and divided as follows-
   - For local Thepade's SBTC n-ary, partitioned vectors into n parts and stored mean values for each part in n different clusters.
   - Ex. For local Thepade's SBTC 3 ary - divided clusters into 3 parts and stored mean values for each part in 3 different clusters.
4. The mean for each cluster is calculated-
   - For Thepade's SBTC n-ary, the mean for the n clusters is calculated and stored in n vectors.
   - Ex. For Thepade's SBTC 3-ary, the mean for the 3 clusters is calculated and stored in 3 vectors.
5. These vectors are considered for the final feature set.

B. Experimentation Environment

The proposed method has been trained and tested on 2 datasets. They are listed below:

a. CASIA 2 Dataset:

CASIA 2 is perhaps the most mainstream accessible dataset in picture fraud discovery. It contains 12,614 shading pictures of size 384 x 256 pixels with BMP, JPEG TIF designs. These pictures are isolated into two subsets: Tampered pictures and Authentic pictures.

There are 7491 pictures in the authentic picture set and 5123 pictures in the tampered picture set. Out of 7491 credible pictures, 7437 pictures are in JPEG design, and the remaining 54 pictures are in BMP design. Likewise, on Tampered pictures, out of 5123 tampered pictures, 2064 pictures are in JPEG design, and the leftover 3059 pictures are in TIFF design.

Fig. 5. Sample Pictures from CASIA 2 Dataset.

b. COMOFOD Dataset:

The dataset contains 10000 shading pictures of size 512 x 512 pixels. It contains pictures of configurations JPEG and PNG. These pictures are partitioned into two subsets: Original pictures and Forged pictures. The first set contains

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5000 pictures, and the Forged set contains 5000 pictures. In this dataset, out of 10000 complete shading pictures, 3600 pictures are in JPEG design, and 6400 pictures are in PNG design.

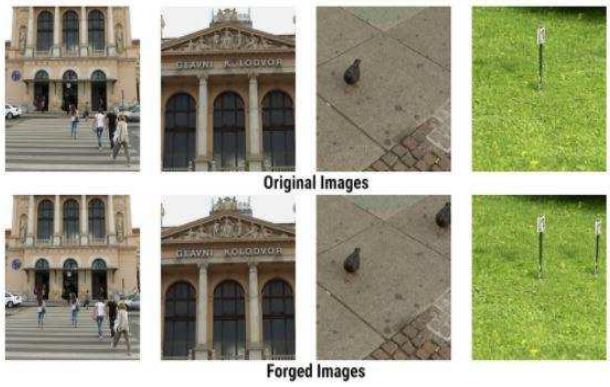

Fig. 6. Sample pictures from the COMOFOD dataset.

In the feature extraction process, there are various upsides of N (for example, N=2,3,4). Global TSBTC N-ary and local TSBTC N-ary are taken and examined to calculate the accuracy of different machine learning classifiers.

The process of global and local feature extraction is done using a python platform. After this process, feature tables are generated, which are fed to different machine learning algorithms. For training and testing of the dataset as well as final results generation, the Weka platform is used.

IV. Result and Discussion

In the following result tables and graphs the terms used include global 2, local 2, global 2 + local 2, global 3, local 3, global 3 + local 3, global 4, local 4, global 4 + local 4. Here, Global n represents Thepade's SBTC n ary on global features, Local n represents the Thepade's SBTC n ary on local features and Global n + Local n represents Thepade's SBTC n ary on fusion of global and local features.

A. CASIA Dataset:

TABLE I

EXECUTION & EVALUATION OF MACHINE LEARNING CLASSIFIERS FOR PROPOSED LOCAL AND GLOBAL THEPADE'S SBTC (TILL 4-ARY) BASED ON IMAGE FORGERY DETECTION EXAMINED ON CASIA DATASET

CASIA DATASET	Random forest	Random tree	Naïve Bayes	Logistic regression	SVM	**Average**
global 2	60.349	55.914	60.626	62.657	61.610	60.231
local 2	59.817	55.168	59.024	62.411	59.428	59.170
global 2 + local2	61.666	55.954	59.508	62.832	61.491	60.290
global 3	62.467	56.723	60.721	63.427	62.776	61.223
local 3	59.307	54.859	60.202	63.968	59.386	59.544
global 3 + local 3	63.189	56.683	60.285	64.617	62.895	61.534
global 4	63.800	57.413	60.610	63.736	63.030	61.718
local 4	59.148	55.137	60.012	64.150	59.386	59.567
global 4 + local 4	63.911	56.739	60.142	65.275	63.117	61.837
Average	61.517	56.066	60.126	63.675	61.458	

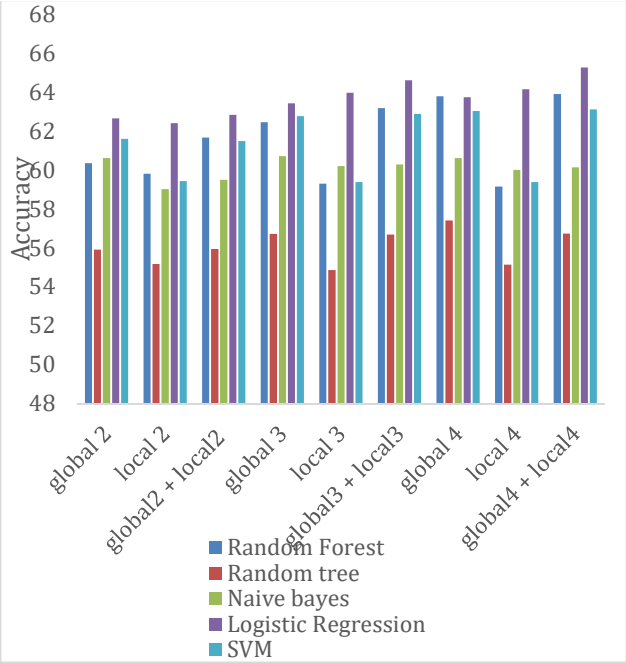

Fig.7. Comparative study of Local +TSBTC n-ary on different ML classifiers at n=2,3,4 on CASIA 2 dataset.

B. COMOFOD Dataset:

TABLE II

EXECUTION & EVALUATION OF MACHINE LEARNING CLASSIFIERS FOR PROPOSED LOCAL AND GLOBAL THEPADE'S SBTC (TILL 4-ARY) BASED ON IMAGE FORGERY DETECTION EXAMINED ON COMOFOD DATASET

COMOFOD DATASET	Random forest	Random tree	Naïve Bayes	Logistic regression	SVM	**Average**
global 2	92.122	91.592	51.316	50.775	50.355	67.232
local 2	91.37	91.200	50.690	50.760	51.050	67.014
global 2 + local 2	92.032	91.542	50.815	49.975	50.675	67.008
global 3	92.283	91.502	52.136	51.326	51.536	67.757
local 3	93.26	92.64	50.62	50.38	51.310	67.642
global 3 + local 3	93.484	92.683	51.126	50.795	50.785	67.774
global 4	92.303	91.642	51.966	51.9868	51.025	67.785
local 4	92.000	91.490	50.750	52.050	50.470	67.352
global 4 + local 4	92.653	91.592	51.116	51.716	49.654	67.346
Average	92.389	91.765	51.170	51.085	50.762	

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Table I shows the accuracy for global features, local features, and fusion of local and global features. For results, five different classifiers are used: Random Forest, Random Tree, Naive Bayes, Logistic, SVM. The table shows that Logistic and Random Forest are the best-performing algorithms having accuracies of 63.6753 and 61.5173, respectively. It is followed by SVM with an accuracy of 61.4582. It is observed from the table that the accuracies are in increasing order of Thepade's SBTC local, Thepade's SBTC global, and Thepade's SBTC on the fusion of local and global features. Fusion of local and global n-ary (for n=4) is giving the best average accuracy of 61.8373. Table II shows the accuracy for global features, local features, and fusion of local and global features. For results, five different classifiers are used: Random Forest, Random Tree, Naive Bayes, Logistic, SVM. The table shows that Random Forest and Random tree are the best-performing algorithms having accuracies of 92.3899 and 91.7650, respectively.

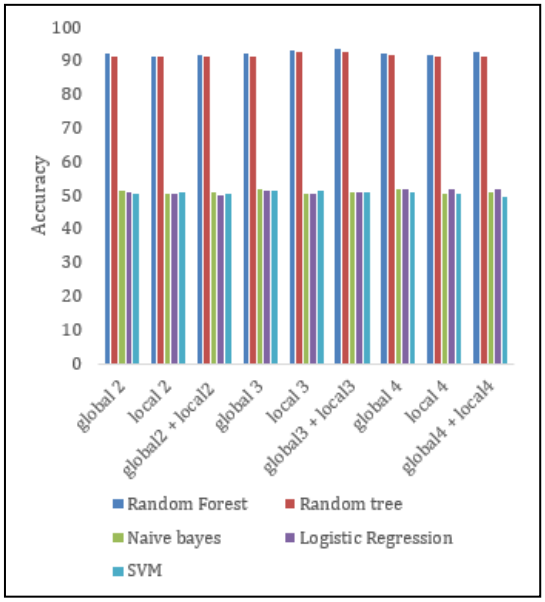

Fig.8. Comparative study of Local +TSBTC n-ary on different ML classifiers at n=2,3,4 on the COMOFOD dataset.

V. Conclusion

Due to progress in computer technology, image forgery can be easily achieved, which can have ill consequences. The misinformation carried and spread by such forged images possesses a significant threat to our society. Using machine learning-based methods to detect image forgery is one of the best possible solutions to this problem.This paper has put forth a special and novel machine learning-based approach for image forgery detection. The proposed strategy utilized N-ary Thepade's SBTC algorithm for global as well as local feature extraction. The features extracted by N-ary Thepade's SBTC Global, Thepade's SBTC Local, and the combination of Thepade's SBTC Local and Global are utilized for preparing and testing machine learning classifiers. The datasets utilized for preparing and testing are CASIA 2 and COMOFOD. The epic methodology essentially centers around the combination of Global and Local Thepade's SBTC. N-ary TSBTC Global (For N=2,3,4) performed better than N-ary TSBTC Local (for N=2,3,4), where the window size for TSBTC local is 3X3. The fusion of TSBTC Global and TSBTC Local performed better than TSBTC Global, where a combination of N-ary TSBTC Local (for N=2,3,4) and N-ary TSBTC Global (for N=2,3,4) is used. After increasing the value of N for local TSBTC (for N=3,4) in the fusion, the model's accuracy started increasing.

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